

Cooper Miles

✉ cooper@psychofx.com in cooper-miles-9b1253376 🐼 Titanium-Lung

Objective

Seeking co-op as a Software Developer. Available: January - December 2027

Education

Rochester Institute of Technology: Rochester, New York

Expected Graduation May 2029

Bachelor of Science in Computer Science

- GPA: 3.800, Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)

Skills

Programming Languages: Java, Python, Javascript, Typescript, HTML, GDScript, C, C#

Technologies: React, Flask, PostgreSQL, Docker, REST APIs, Angular, Gunicorn

Tools: Git, Openshift, Fusion360, Blender

Projects

Archipelago Hosting Service | Personal (current) | github.com/Titanium-Lung/archipelago-csh

Developed a full-stack web application using React and Flask to create and manage multiplayer Archipelago game servers.

- Deployed and configured the application on OpenShift, creating application deployments and routes, and connecting the application to a PostgreSQL database hosted on CSH servers.
- Designed a PostgreSQL-backed persistence layer for storing active game data and restoring hosted rooms after application restarts, allowing rooms to shut down after inactivity and be restored as needed.
- Designed a process-management architecture to coordinate Archipelago server subprocesses across multiple Gunicorn workers, using TCP and JSON messages to communicate with a centralized process manager.

Mock Ufund Website | Class (2025)

Developed a full-stack web application with a Java REST API backend and Angular frontend as part of a four-person team

- Applied object-oriented design principles, classes, and interfaces to create modular and maintainable backend components.
- Collaborated through Git to integrate backend and frontend components and coordinate development across the team.

Basketpong | Personal (2025) | github.com/Titanium-Lung/basketpong

Developed a 2D multiplayer game in Godot/GDScript, combining basketball-style gameplay with Pong mechanics.

- Implemented collision detection, ball physics, player movement, and scoring mechanics to create responsive gameplay.
- Designed and implemented a multi-page user interface for navigating menus and gameplay.

Work Experience

Undergraduate Research Assistant

Rochester, NY

Rochester Institute of Technology

Aug 2026 – Present

- Researching papers related to using LLMs to improve security of protected healthcare information
- Completed training regarding dealing with data collected from human subjects
- **Skills Used:** Teamwork, Research

Activities

Computer Science House

Rochester, NY

Learning community supporting member growth in various computing disciplines

Aug 2024 - Present

House History Director

Jun 2026 – Present

- Supervising over House History operations
- Maintaining, upholding, and promoting house traditions
- Contacting companies in order to find a venue for house's annual formal dinner
- Discussing and voting on issues presented to E-Board
- **Skills Used:** Communication, professional email